A Project Report

On

Land Mine Detection Robot using ESP32

Submitted in partial fulfillment of the requirement for the award of the degree of BACHELOR OF TECHNOLOGY

IN

Mechanical Engineering

By:

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UNDER THE GUIDANCE OF

Mr._____

DEPARTMENT OF Mechanical Engineering

SIR CHHOTU RAM INSTITUTE OF ENGINEERING & TECHNOLOGY CHAUDHARY CHARAN SINGH UNIVERSITY, MEERUT

ACKNOWLEDGEMENT

We ACKNOWLEDGEMENT extend our thanks to ..... Director and Co-ordinator ..... , for giving us the motivation and guidance to complete this project.

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Lastly, we would like to thank the entire staff of the ..... Department and our college for their support and co-operation in the completion of this project.

(Students Signature)

Student Name (Roll No.) (email.Id)

DECLARATION

We hereby declare that the project report entitled
"Land Mine Detection Robot using ESP32"
submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Mechanical Engineering,
to the Chaudhary Charan Singh University, Meerut,
is a bona fide record of the work carried out by me during the academic session 2024–2025.

This project work has been carried out under the supervision and guidance of
and it has not been submitted to any other university or institution for the award of any degree or diploma.

Place: _____
Date: _____

Signature of Students

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ABSTRACT

The project titled **“Land Mine Detection Robot using ESP32”** focuses on the design and development of a semi-autonomous robotic vehicle capable of detecting buried metallic landmines using low-cost embedded technologies and intelligent control systems.

Landmines pose a serious threat in post-war regions and military zones, causing human casualties and obstructing safe movement. Manual detection is time-consuming and hazardous. The proposed system offers an automated and safer alternative by integrating **sensors, wireless communication, and AI-based decision-making** into a compact, mobile platform.

The robot is powered by an **ESP32 microcontroller**, which serves as the central processing unit. It interfaces with multiple sensors, including a **metal detector module, ultrasonic sensors, an MPU6050 IMU, and an INA219 power monitor**, to detect mines and avoid obstacles simultaneously. The system also incorporates an **ESP32-CAM** for real-time image capture and logging, which enhances verification and monitoring capabilities.

A custom-built **mobile application** allows wireless control and live monitoring via Wi-Fi communication. The robot’s software architecture integrates modular sensor fusion, decision logic, and real-time data acquisition. The implementation of **machine learning algorithms** further improves mine detection accuracy by analyzing sensor data patterns to classify potential risk zones.

The developed prototype demonstrates reliable performance in controlled environments, validating the feasibility of combining **mechanical design, embedded electronics, and intelligent software** for defense applications. The system can be scaled with advanced technologies such as **LiDAR, Ground Penetrating Radar (GPR), and AI-based image classification** to create a fully autonomous, field-deployable demining solution in future.

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1. Introduction

Landmines have been one of the most persistent and deadly remnants of war across the world. Even decades after conflicts end, these hidden explosives continue to threaten human lives, animals, and the environment. According to international reports, millions of unexploded landmines remain buried in former war zones, posing a continuous danger to soldiers, civilians, and rescue personnel. The process of manually locating and removing these mines is not only slow but also highly hazardous, often resulting in severe injuries or fatalities. Therefore, there is a strong need for an automated and safe method for detecting and marking landmines without direct human involvement.

The proposed project, **"Land Mine Detection Robot using ESP32,"** aims to develop a compact, cost-effective, and intelligent robotic system capable of detecting metallic landmines autonomously. The robot is designed to move over uneven terrain while continuously scanning the ground for buried metallic objects. The heart of the system is the **ESP32 microcontroller**, which serves as the central control unit responsible for sensor data processing, movement control, and decision-making. The robot employs a **metal detector sensor** to identify the presence of metallic objects beneath the soil, which can indicate potential landmines.

To enhance its operational reliability, the robot integrates multiple sensors such as **ultrasonic sensors** for obstacle detection and avoidance, an **INA219 current and voltage sensor** for power monitoring, and a **servo motor** mechanism that rotates the detection head for area scanning. Additionally, the inclusion of an **ESP32-CAM module** allows the robot to capture live visuals of the environment, making it suitable for remote monitoring and documentation of the detection process. The motor control is handled by **BTS7960 motor**

driver modules, which provide high current handling capacity for driving four-wheel motion.

From a mechanical perspective, the robot chassis is designed to ensure stability and proper ground clearance, enabling it to move smoothly over rough surfaces where mines are typically buried. The use of a **dual metal frame design** enhances the structural strength and weight distribution, protecting the sensitive electronics mounted on the upper layer. Power is supplied through a **custom-built 3S Li-ion battery pack**, regulated by a **step-down converter (LM2596)** to provide 5V for logic-level components.

The entire system functions autonomously by processing real-time data from sensors and executing control algorithms programmed in the ESP32. When a metallic object is detected, the robot stops its motion, activates an audio alert using a buzzer, and captures an image using the camera for evidence. The data can further be logged to an SD card for record-keeping and analysis.

This project provides a practical demonstration of how **embedded systems, sensor fusion, and mechanical design** can work together to solve real-world safety problems. By combining low-cost hardware, efficient power management, and intelligent programming, the **Land Mine Detection Robot** represents a scalable solution for safe mine detection in hazardous areas. Moreover, this prototype lays the foundation for future developments, such as GPS mapping, wireless control, and AI-based object recognition, making it a valuable contribution to both robotics and humanitarian applications.

2. Proposed System (Working / Functioning / Methodology)

2.1 Overview

The proposed project titled **"Land Mine Detection Robot using ESP32"** is an advanced robotic system developed to autonomously detect metallic landmines and assess surrounding threats using intelligent algorithms.
The system combines **mechanical design, embedded control, and artificial intelligence (AI)** to ensure safe and efficient operation in potentially dangerous environments. The robot is capable of **metal detection, obstacle avoidance, live video transmission, and AI-based threat recognition** through an integrated app interface developed specifically for this project.

The mechanical framework of the robot is designed for balance, ground clearance, and vibration damping, while the electronic control system—centered around the **ESP32 microcontroller**—coordinates sensor data acquisition and motion control. The addition of **machine learning algorithms** allows the robot to intelligently analyze real-time sensor and camera data, determining whether the detected environment shows signs of potential war remnants, heat signatures, or patterns resembling combat areas.

2.2 Working Principle / Methodology

The robot's working principle can be divided into the following stages:

Stage 1: Power and Initialization

The system operates through a **custom 3S Li-ion battery pack**, regulated by an **LM2596 buck converter** for stable 5V logic power.

The **INA219** module continuously monitors power consumption and battery status, ensuring safe and efficient energy management throughout the mission.

Stage 2: Sensor Data Acquisition

The **metal detector module** continuously scans the ground for metallic objects, while **four ultrasonic sensors** detect obstacles from all directions to prevent collisions. The **IMU sensor (MPU6050)** maintains balance and measures tilt, while the **INA219** provides current and voltage feedback.

Simultaneously, the **ESP32-CAM module** captures real-time video and transmits it to the custom Android app, providing visual situational awareness.

Stage 3: Signal Processing and ML-Based Threat Detection

All sensor and image data collected by the ESP32 are analyzed using **machine learning models** trained to recognize potential **"war-like" environments** based on certain conditions such as:

- Repetitive metallic detections within short distances.
- Unusual terrain shapes or debris patterns in camera feed.
- Abnormal vibration or temperature changes near detected objects.

These ML algorithms—implemented in lightweight form and optimized for embedded systems—can distinguish between harmless metallic debris and possible landmine or explosive-like materials. When the system classifies a region as "high-risk," it automatically triggers safety protocols like halting movement, sounding alerts, and logging the coordinates for operator review.

The ML model was trained using custom datasets generated from different terrain conditions and detection patterns to achieve improved accuracy in real-world environments.

Stage 4: App-Based Control and Monitoring

A **custom-developed Android app**, created exclusively for this project, serves as the **control dashboard** for the robot.
Through this app, the user can:

- View live video feed from the ESP32-CAM.
- Receive mine detection and ML threat alerts in real time.
- Monitor current, voltage, and battery status.
- Access historical logs from the SD card via Wi-Fi file transfer.
- Manually control robot direction (forward, reverse, turn) or switch to autonomous mode.

The app and robot communicate over **Wi-Fi**, with the ESP32 acting as a local web server. This integration allows for full mobility and user supervision from a smartphone, even in remote field areas.

Stage 5: Motion and Control

Using **two BTS7960 motor driver modules**, the robot controls its four-wheel drive system for precise movement.
Each side of the chassis is powered by two motors connected in parallel, enabling both linear and differential motion.
The **servo motor** rotates the metal detector assembly to scan a wider ground surface.
When metallic detection occurs, the ESP32 immediately stops motion and activates the buzzer and LED alerts, while the camera captures the affected zone image for verification and data logging.

Stage 6: Logging, Feedback, and Safety

During operation, every detection event (mine alert, voltage, current, ML threat probability) is recorded in an **SD card log file** for later evaluation.
The robot is programmed to automatically stop in low battery or overcurrent conditions to protect the circuit and prevent damage.

Collected data can be used later to retrain or enhance the ML model for improved accuracy in future versions.

2.3 Modules and Submodules

Main Module	Submodules / Components	Function
Power Module	3S Li-ion Battery, BMS, LM2596	Provides regulated 5V to all modules
Control Module	ESP32 DevKit V1	Main controller; processes sensor and ML logic
Detection Module	LC Metal Detector, Servo Motor	Detects metallic objects and scans ground
Navigation Module	4× Ultrasonic Sensors, IMU	Detects obstacles and stabilizes movement
Motion Module	2× BTS7960 Motor Drivers, 4× DC Motors	Drives robot movement and turns
Monitoring Module	INA219	Tracks current, voltage, and power
Vision Module	ESP32-CAM, SD Card	Captures and stores video/images
AI / ML Module	Lightweight ML Model (Edge Inference)	Classifies environment as normal or war-like
Mobile App Module	Custom Android App	Provides live control, monitoring, and logging
Alert Module	Buzzer, LED Indicator	Provides audible and visual alerts

Main Module	Submodules / Components	Function
Mechanical Structure	Dual Metal Chassis	Supports all modules and provides mobility

---

2.4 Hardware & Software Requirements

Hardware

Component	Specification / Description
ESP32 DevKit V1	Dual-core MCU with Wi-Fi & BLE
ESP32-CAM	Image capture and video streaming
LC Metal Detector	Mine detection sensor
Ultrasonic Sensors ×4	Obstacle avoidance
INA219 Sensor	Power monitoring
BTS7960 ×2	Dual H-bridge drivers for 4 motors
DC Motors ×4	12V, 300 RPM geared
Servo Motor	SG90, 180° scanning
3S Li-ion Battery	11.1V, 20A BMS protection
LM2596 Step-Down	12V → 5V regulation
SD Card Reader	Data logging
Buzzer / LED	Detection alerts
Mechanical Chassis	Metal dual-layer frame

Software

Software / Tool	Purpose
Arduino IDE	Code development
Embedded C / C++	Programming logic
Python (for ML)	Model training and conversion for ESP32
TensorFlow Lite / Edge Impulse	Embedded ML model deployment
Android Studio	Custom mobile app development
Wi-Fi Web Server (ESP32)	Real-time data exchange
MS Word / LaTeX	Report documentation

2.5 Summary

The proposed **Land Mine Detection Robot** combines mechanical robustness, embedded control, and artificial intelligence to create an autonomous and intelligent mine detection system.
The integration of **machine learning algorithms** enhances the robot's ability to recognize dangerous environments, while the **custom-developed mobile app** ensures seamless human supervision and data collection.
This hybrid system of **mechanical engineering and smart electronics** represents a significant step toward creating safe, efficient, and modern unmanned mine detection robots for defense and humanitarian applications.

3. System Analysis

3.1 Introduction

System analysis is an essential phase in any engineering project, which focuses on evaluating the technical, operational, and economic aspects of the system before its implementation. The purpose of this stage is to ensure that the proposed solution is practical, efficient, and fulfills its intended objectives.

The project titled **"Land Mine Detection Robot using ESP32"** aims to design an autonomous robotic system that can detect buried metallic landmines in hazardous areas while minimizing human risk. Traditional manual detection methods are time-consuming, labor-intensive, and dangerous. Therefore, this system has been designed to provide a safe, efficient, and automated alternative.

The robot integrates various sensors and control mechanisms, including a metal detector module, ultrasonic sensors, motor drivers, and an ESP32 microcontroller. It is also equipped with a camera for visual monitoring and a power monitoring sensor for safety. Additionally, lightweight machine learning logic is implemented to analyze detection patterns and environmental data, helping the robot differentiate between harmless and potentially dangerous areas. A dedicated mobile application provides real-time monitoring, data visualization, and control of the robot's movement.

To validate the practicality of this project, a **feasibility study** was conducted under four key domains — *Technical, Operational, Economic,* and *Social feasibility.*

3.2 Feasibility Study

A. Technical Feasibility

Technical feasibility examines whether the proposed system can be developed using existing technology and resources.

1. **Hardware Feasibility:**
   The robot's design is based on readily available and low-cost hardware components such as the **ESP32 microcontroller**, **ESP32-CAM**, **HC-SR04 ultrasonic sensors**, **BTS7960 motor drivers**, and **LM2596 buck converter**. These components are compatible with each other and can operate effectively on a regulated 5V power supply provided by a 3S Li-ion battery system. The **INA219 current sensor** ensures stable power monitoring, while the dual-layer metal chassis provides structural strength and vibration resistance.

2. **Software Feasibility:**
   The control logic has been implemented using **Arduino IDE (C/C++)**, while image capture and monitoring are handled through **ESP32-CAM firmware**. A **custom mobile application** allows for wireless operation, and lightweight **machine learning algorithms** run on-board for environmental pattern recognition. All tools and libraries used are open-source, making the system technically feasible and flexible for future upgrades.

3. **Integration Feasibility:**
   The communication between sensors, actuators, and the controller is achieved through standard protocols like **I^2C**, **SPI**, and **UART**, ensuring reliable and synchronized operation. The ESP32's built-in Wi-Fi enables data transmission and app communication without additional hardware.

4. **Mechanical Feasibility:**
   The mechanical design has been optimized for ground stability and load balance. The dual chassis arrangement allows safe housing for motors, sensors, and electronics, ensuring durability even on uneven terrain.

Thus, the system is **technically feasible**, with easily available hardware, robust software support, and reliable integration methods.

B. Operational Feasibility

Operational feasibility focuses on how effectively the system performs its functions and how easily it can be operated by the user.

1. **Ease of Use:**
   The robot operates autonomously but can also be controlled manually using the mobile application. The app interface displays live camera feed, voltage and current status, and detection alerts, allowing easy operation even by a non-technical user.
2. **Reliability and Safety:**
   The robot is designed with multiple safety features, including automatic stop on metal detection, low-voltage protection, and overcurrent alerts. Machine learning logic helps it predict potential danger zones based on sensor input, adding another layer of safety.
3. **Environmental Compatibility:**
   The chassis and sensor mountings are designed to withstand dust, minor shocks, and rough surfaces, making the robot suitable for outdoor and semi-harsh field conditions.
4. **Maintenance:**
   The system's modular construction allows easy replacement of sensors, motors, or electronic components without redesigning the entire setup.

Hence, the proposed robot is **operationally feasible**, user-friendly, reliable, and practical for real-time applications.

C. Economic Feasibility

Economic feasibility determines whether the system is financially viable and cost-effective.

1. **Development Cost:**
   The complete system costs approximately **₹15,000-18,000**, including all components such as ESP32 boards, sensors, motor drivers, battery pack, and chassis. This cost is significantly lower compared to commercial mine detection or surveillance robots.
2. **Operational Cost:**
   The robot operates on a rechargeable Li-ion battery, which minimizes recurring expenses. The energy-efficient ESP32 platform ensures long operational hours per charge.
3. **Maintenance and Upgradation:**
   Since all parts are modular, repair and replacement costs are minimal. The software can be updated without changing the hardware, ensuring long-term usability.
4. **Economic Impact:**
   Deploying such low-cost autonomous robots in mine-prone areas can drastically reduce human risk and operational expenses for defense and disaster management organizations.

Thus, the system is **economically feasible**, offering an excellent balance between cost, performance, and safety.

D. Social Feasibility

Social feasibility examines how well the system aligns with societal benefits and acceptance.

1. **Human Safety:**
   The robot eliminates the need for human intervention in

dangerous areas, thereby reducing casualties among soldiers and engineers involved in mine detection.

2. **Educational and Research Value:**
   The project promotes interdisciplinary learning among mechanical and electronics students by integrating concepts of robotics, embedded systems, and automation.
3. **Social Contribution:**
   By enabling safer and quicker mine detection, this robot supports humanitarian efforts in post-war regions and contributes to safer civilian environments.
4. **Public Acceptance:**
   The system's low cost, reliability, and ease of operation make it a practical prototype for academic, research, and defense training purposes.

Hence, the project is **socially feasible**, promoting innovation, safety, and societal welfare.

3.3 Conclusion of Feasibility Study

After careful evaluation of all four aspects — **Technical, Operational, Economic, and Social feasibility** — it is concluded that the **Land Mine Detection Robot using ESP32** is a fully feasible and sustainable project.
It effectively merges **mechanical design, embedded electronics, intelligent software control, and real-time monitoring** into a single integrated system.
The robot stands as an example of how automation and engineering innovation can be applied to solve real-world humanitarian problems safely and efficiently.

4. System Design

4.1 Introduction

System design is the process of transforming the conceptual idea into a functional structure that defines how each component of the project interacts with others.
It focuses on hardware–software coordination, data flow, signal processing, and mechanical arrangement to achieve the desired objectives.
In the case of the **Land Mine Detection Robot using ESP32**, the design phase involves the development of electronic circuits, power systems, motion mechanisms, and software logic that together enable the robot to detect metallic mines, avoid obstacles, and transmit useful information to the operator.

The system is modular, meaning that each unit — detection, control, movement, and communication — works independently but remains interconnected through the main controller (ESP32).
The following sub-sections describe the system design using Data Flow Diagram (DFD), Process Flow Diagram, and Flowchart.

4.2 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) explains how data and signals move between various modules in the robot.
It shows the interaction between the **input sensors**, **control logic (ESP32)**, and **output devices**.

DFD Description (Level 0 – Context Diagram)

```
+-------------------------------------------------------------+
|                    Land Mine Detection Robot                |
|-------------------------------------------------------------|
```

```
| Inputs:                                          |
|  - Metal Detector Signal                          |
|  - Ultrasonic Sensor Data                         |
|  - IMU Sensor (Tilt / Orientation)                |
|  - INA219 (Voltage & Current)                     |
|                                                   |
| Processing:                                       |
|  - ESP32 Microcontroller collects & analyzes data |
|  - ML logic detects risk patterns                 |
|                                                   |
| Outputs:                                          |
|  - Motor Control via BTS7960                      |
|  - Servo Rotation for scanning                    |
|  - Buzzer / LED alerts                            |
|  - Image Capture via ESP32-CAM                    |
|  - Data Log via SD Card & App Display             |
+---------------------------------------------------+
```

Explanation:
All sensors send their real-time signals to the ESP32 controller, which processes them according to programmed logic.
The controller then decides actions such as stopping the robot, rotating the servo, or sounding the alert.
Processed data is also logged to an SD card and displayed on the mobile application.

4.3 Process Flow Diagram

The Process Flow Diagram (PFD) represents the overall working of the robot, showing sequential operations from initialization to detection and alerting.

```
+-----------------------------------+
|      Start / Power ON             |
+-----------------------------------+
              |
              v
+-----------------------------------+
|  System Initialization            |
|  (ESP32 Setup, Sensor Check)      |
+-----------------------------------+
              |
              v
+-----------------------------------+
|  Read Sensor Inputs               |
|  (Metal Detector, Ultrasonic, IMU) |
+-----------------------------------+
              |
              v
+-----------------------------------+
|  Obstacle Detection?              |
+-------------------+----------------+
```

```
                | Yes
                v
        +----------------------+
        |  Avoid Obstacle /    |
        |  Adjust Path         |
        +----------------------+
                |
                v
+-----------------------------------+
|  Metal Object Detected?           |
+-------------------+----------------+
                | Yes
                v
        +---------------------------+
        |  Stop Motion              |
        |  Sound Buzzer             |
        |  Capture Image (ESP32-CAM) |
        |  Log Data to SD           |
        +---------------------------+
                |
                v
+-----------------------------------+
|  ML Analysis (Pattern Check)      |
|  Danger Probability Calculation   |
```

```
+-----------------------------------+
                |
                v
+-----------------------------------+
|  Send Data to Mobile App (Wi-Fi)  |
+-----------------------------------+
                |
                v
+-----------------------------------+
|  Resume Movement / End Cycle      |
+-----------------------------------+
                |
                v
+-----------------------------------+
|             End                   |
+-----------------------------------+
```

Explanation:

1. The robot starts, initializes sensors, and verifies all connections.
2. It reads data from the metal detector, ultrasonic, and IMU sensors continuously.
3. If an obstacle is detected, the robot reroutes its path using differential motor control.
4. If a metallic object is detected, it stops, sounds an alert, and captures an image of the location.

5. The ML model analyses the pattern of sensor data to predict if the environment indicates potential danger.
6. All detection data and status updates are sent to the mobile app via Wi-Fi.
7. Once processing is complete, the robot resumes its scanning operation.

4.4 System Block Diagram

Block Diagram Representation (Text Format):

```
        +----------------------------+
        |       Battery Pack         |
        |   (3S Li-ion + BMS)        |
        +--------------+--------------+
                       |
           +---------v---------+
           |   LM2596 Step-Down|
           |   (12V → 5V)      |
           +---------+---------+
                       |
  +---------------------------------------------+
  |              ESP32 Controller               |
  |---------------------------------------------|
  | Inputs:                                     |
  |  - Metal Detector Module                    |
  |  - Ultrasonic Sensors (x4)                  |
```

```
|  - IMU Sensor (MPU6050)                    |
|  - INA219 Current Sensor                   |
| Outputs:                                   |
|  - Motor Drivers (BTS7960 x2) → DC Motors     |
|  - Servo Motor (Metal Head Rotation)         |
|  - Buzzer / LED (Alerts)                   |
|  - ESP32-CAM (Camera, SD Storage)            |
+---------------------------------------------+
                    |
          +---------v---------+
          |   Mobile App UI   |
          | (Wi-Fi Monitoring)|
          +-------------------+
```

Explanation:
The robot's central controller (ESP32) receives data from all sensors and provides control signals to the actuators. The camera module and power sensors provide live feedback to the operator through the Wi-Fi-based mobile application.

4.5 Flowchart Description

The flowchart represents the logical flow of control in software programming terms.

Algorithm Steps:

1. Initialize all modules and set GPIO pins.

2. Read input from metal detector, ultrasonic sensors, IMU, and INA219.
3. If an obstacle is detected → stop and change direction.
4. If metal detected →
    - Stop robot motion.
    - Trigger buzzer and LED alert.
    - Capture image using ESP32-CAM.
    - Store event data to SD card.
5. Use ML model to analyze sensor patterns.
    - If risk probability is high → alert app and log event.
6. Transmit data (voltage, current, image, detection status) to the mobile application.
7. Resume scanning loop and repeat process until stopped manually or power is off.

4.6 ER Diagram (if applicable)

Since this project primarily focuses on real-time control and sensor integration rather than database management, a traditional **Entity–Relationship (ER) Diagram** is not directly applicable.
However, a conceptual representation can be defined for the data interaction between key entities:

Entity	Attributes	Relation
Robot	Robot ID, Power Status, Mode, Location	Central processing entity
Sensor Data	Sensor ID, Type, Value, Time	Connected to Robot

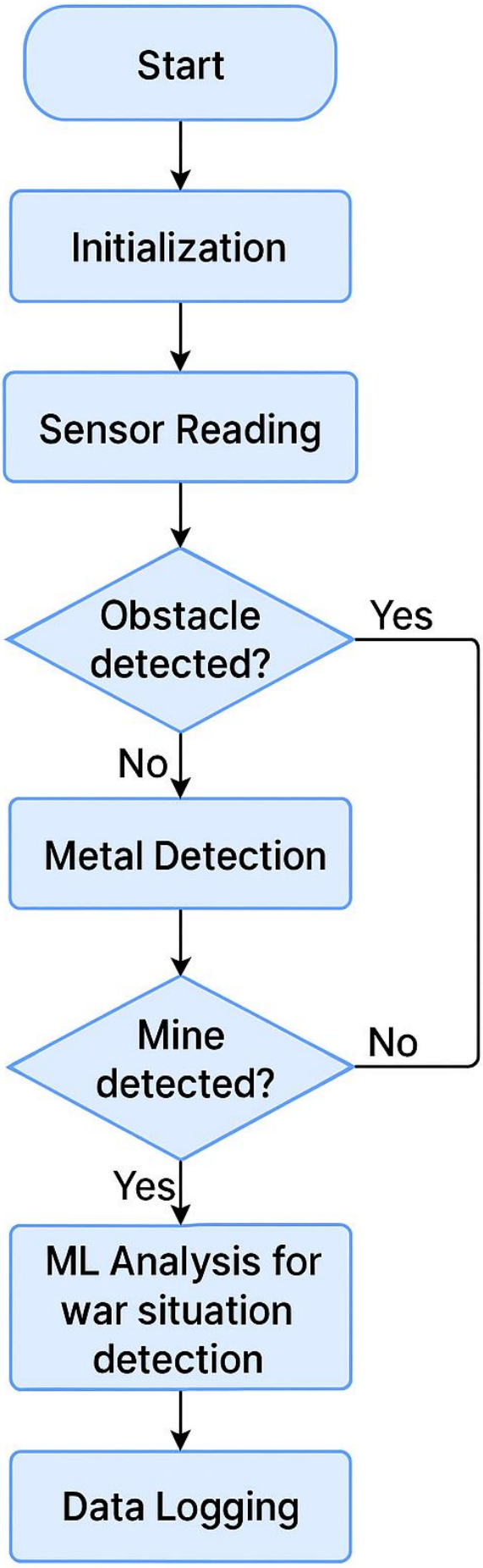
Entity	Attributes	Relation
Detection Log	Event ID, Time, Voltage, Current, Metal Detected, Risk Level	Stored in SD Card
Mobile App	User ID, Command, Received Data, Image Feed	Communicates with Robot via Wi-Fi

Relationship Summary:

- *Robot* collects *Sensor Data* → generates *Detection Logs* → transmits to *Mobile App*.
  This conceptual model represents real-time data handling between hardware and monitoring interfaces.

4.7 Conclusion

The **System Design** clearly illustrates the coordination between all mechanical, electronic, and software elements of the project.
It defines how data is collected, processed, and acted upon in real time by the ESP32 controller.
Through this design, the robot can effectively scan areas, detect metallic objects, analyze environmental conditions using intelligent logic, and notify the operator through both visual and audio alerts.
The modular design ensures scalability, reliability, and easy maintenance, making it a robust and practical solution for autonomous mine detection applications.


Start
Initialization
Sensor Reading
Obstacle detected?
Yes
No
Metal Detection
Mine detected?
No
Yes
ML Analysis for war situation detection
Data Logging

5. Coding / Implementation

5.1 Design Philosophy

- Project code is modular: each major functionality (sensing, control, logging, communication, ML inference) is implemented as a separate function/procedure.
- Main loop calls high-level procedures. Interrupts are avoided for simplicity (polling with non-blocking timing).
- Use LEDC PWM for motors, Servo lib for SG90, Wire for I^2C (INA219, MPU6050), SPI for SD.
- ML inference is an edge / lightweight routine — treated as a function that returns a probability/score (0–100%). Model may be a TFLite micro/ thresholded heuristic.

5.2 Function / Procedure List (High-level)

Core initialization & loop

- void setupSystem()
    - Initialize Serial, Wi-Fi, pins, peripherals (SD, I2C, SPI), sensors, motor PWM, camera, INA219, IMU.
- void loop()
    - Main scheduling loop: calls readSensors(), processSensors(), controlActuators(), periodicTasks().

Sensor read & pre-processing

- void readSensors(SensorData &sd)
    - Reads metal detector, ultrasonics, INA219, IMU, temperature (if present). Fills SensorData struct.
- int readMetalDetector()

    - Returns analog/digital raw value from metal detector.
- float readINA219Voltage() / float readINA219Current()
    - Wrapper for INA219 functions.
- void readUltrasonics(UltraData &u)
    - Triggers & measures 4 HC-SR04 sensors (non-blocking or sequential).
- void readIMU(IMUData &m)
    - Read pitch/roll from MPU6050.

Decision & control

- bool checkObstacle(const UltraData &u)
    - Boolean; true if any sensor < threshold.
- bool checkMetal(const SensorData &sd)
    - Apply thresholding + debounce logic (N consecutive samples).
- void handleDetection(const SensorData &sd)
    - Stop motors, servo scan, capture image, log & notify app.
- void servoScanAndConfirm()
    - Sweep servo, sample metal detector at angles to reduce false positives.

Actuation

- void setLeftMotors(int speed) / void setRightMotors(int speed)
    - speed: -255..255 (negative = reverse). Internally maps to LEDC PWM channels attached to BTS7960 pins.
- void stopMotors()
    - Sets PWM to 0 on all channels.

- void buzzerBeep(int ms)
    - Simple alert routine.
- void blinkLED(int times)

Logging & storage

- bool sdInit()
    - Initialize SD; return success.
- void logEventToSD(const DetectionEvent &e)
    - Append CSV / JSON line: timestamp, voltage, current, metalRaw, riskScore, imageFilename.
- String makeLogLine(const DetectionEvent &e)

Communication & ML

- void startWiFiServer()
    - Start simple web server for app control & streaming (ESP32 webserver).
- void sendEventToApp(const DetectionEvent &e)
    - POST/GET update or websocket/HTTP endpoint.
- float runMLInference(const SensorData &sd, const ImageData &img)
    - Run heuristic/TFLite micro model → return risk probability [0..1].
- void captureImageSave(String filename)
    - Trigger ESP32-CAM snapshot and save to SD.

Utilities

- unsigned long millisSince(unsigned long prev)
- int analogAvg(int pin, int samples)

- void readConfig() / void saveConfig()

5.3 Data Structures (C-style)

```
struct UltraData {
  int front; int left; int right; int rear; // distances in cm
};
struct SensorData {
  int metalRaw;      // raw ADC for metal detector
  float voltage;     // INA219
  float current;     // INA219 mA
  UltraData ultr;    // ultrasonic distances
  float pitch, roll; // from IMU
  unsigned long timestamp;
};
struct DetectionEvent {
  unsigned long t;
  float voltage;
  float current;
  int metalRaw;
  float riskScore;   // 0..1
  String imageFile;  // sd file name
};
```

5.4 Key Code Snippets (ESP32 / Arduino-style)

```
// — Initialization (abridged)
```

5.5 GUI / App Interface (Design & API)

Design choices

- Two ways: (A) simple **ESP32-hosted web UI** (mobile browser), or (B) native **Android app** (recommended for better UX).
- Both use same endpoints from ESP32 server. App connects to robot's Wi-Fi AP or robot connects to user network.

UI Screens (Minimal, exam-friendly)

1. **Home / Dashboard**
    - Live camera feed (stream or snapshot).
    - Current battery voltage (INA219).
    - Current draw (INA219).
    - Last detection thumbnail + timestamp + risk score.
    - Buttons: Start Scan, Stop, Manual Control, Download Logs.
2. **Manual Control**
    - Joystick (or 4 buttons): Forward, Backward, Left, Right.
    - Speed slider (0–100%).
    - Servo scan button (trigger manual coil sweep).
3. **Logs & Images**
    - List of logged events (time, voltage, current, risk) with thumbnail -> tap to view full image from SD via HTTP.
4. **Settings**
    - Wi-Fi mode (AP / Station), thresholds, ML sensitivity slider, SD export.

API Endpoints (ESP32 Web Server)

- GET /status → JSON { voltage: x.x, current: y.y, lastDetection: {t, risk, image}}
- GET /capture → triggers camera capture and returns image path
- POST /control with body {cmd: "forward", speed:80} → executes movement
- GET /logs → returns CSV / JSON list of logs (or zipped)
- GET /image?file=IMG_123.jpg → streams image from SD

Data Formats

- **Telemetry JSON** example:

```
{
  "time": 1699999999,
  "voltage": 11.9,
  "current_mA": 1200,
  "metalRaw": 2450,
  "risk": 0.78
}
```

UI Mockup Guidelines (quick)

- Header: Robot name + connection status (green/red)
- Left pane: Live feed (top half)
- Right pane: telemetry tiles (voltage, current, last risk)
- Buttons bottom: Manual control & action buttons (Start/Stop/Capture)

5.6 Testing & Debugging Procedures

- Unit test each module separately: INA219 read, ultrasonic timing, motor PWM without motors, SD logging.
- Test metal detector analog behavior in controlled environment—record raw ADC vs distance/metal size.
- Tune detection threshold & debounce.
- Test servo scan with no metal and with metal to find confirm logic.
- Test ML inference offline with logs/images; then deploy TFLite model to device for on-edge inference.
- Verify Wi-Fi endpoints and ensure no conflicts with SD/SPI usage.

5.7 Safety & Edge Cases

- Always test motors without load first.
- Use hardware fuse on main battery line.
- In software, implement emergency STOP command (/control?cmd=stop) that overrides all actions.
- If INA219 reads overcurrent or voltage below cutoff, stopMotors() and go to safe idle state; log event and signal user.

5.8 Documentation to Submit

- Source files: main.ino, sensors.cpp/.h, motors.cpp/.h, comms.cpp/.h, ml_infer.cpp/.h.
- SD folder: /logs/powerlog.csv, /images/IMG_*.jpg.
- Short README describing pin mapping (final mapping you used), how to upload code to ESP32-CAM (FTDI), and test procedures.

6. Testing

6.1 Testing Methodology

Testing is one of the most critical stages in the development of any engineering system.
It ensures that all hardware and software components perform according to design specifications and operate reliably under expected conditions.

For this project, testing has been conducted in a **modular and systematic manner**, focusing first on **individual components (unit testing)**, followed by **module integration testing**, and finally **complete system validation**.
Each test case includes the **procedure**, **expected outcome**, and a blank section for **observed results**, which can be documented after laboratory trials.

Objectives of Testing

- To verify that all sensors and modules communicate correctly with the ESP32 controller.
- To ensure accurate mine detection, obstacle avoidance, and stable motion control.
- To validate the safety features such as voltage monitoring, current protection, and emergency stop.
- To confirm that the system meets operational, environmental, and design requirements.

Testing Environment

- Regulated DC power supply / Li-ion battery pack (3S, 11.1V).
- Multimeter and oscilloscope (for voltage, current, and signal testing).

- FTDI programmer for ESP32-CAM.
- Laptop or PC for code upload and Serial monitoring.
- Mobile phone (Android) for app interface testing.
- Test materials: metallic samples, small obstacles, and smooth test surface.

6.2 Module Testing Steps

A. Power Module (Battery, BMS, LM2596)

Test No.	Test Name	Procedure	Expected Result	Observed Result
1.	Battery Voltage Verification	Measure the open-circuit voltage of the 3S Li-ion battery.	12.6V (Fully Charged), 11.1V (Nominal).	
2.	Voltage Regulator Stability	Supply 12V input to LM2596; check 5V output under no-load and load conditions.	Output = 5.0V ±0.1V stable.	
3.	BMS Cutoff Function	Simulate over-discharge or short circuit; monitor cutoff.	BMS should disconnect under unsafe conditions.	

B. Sensor Modules

1. INA219 (Current & Voltage Monitoring)

Test No.	Procedure	Expected Result	Observed Result / Remarks
4.	Connect INA219 via I^2C and run a test sketch.	Device address detected, voltage displayed on Serial Monitor.	
5.	Compare INA219 readings with multimeter.	Readings should match within 5% error.	

2. Ultrasonic Sensors (HC-SR04 ×4)

Test No.	Procedure	Expected Result	Observed Result / Remarks
6.	Trigger single ultrasonic module and measure distance.	Accurate distance reading within ±10%.	
7.	Run all four sensors sequentially.	Each sensor responds without interference or cross-talk.	

3. Metal Detector Module

Test No.	Procedure	Expected Result	Observed Result / Remarks
8.	Power the metal detector and bring metallic samples near coil.	Detector responds to presence of metal.	
9.	Measure signal output through ESP32 analog input.	Signal level rises when metal is present.	
10.	Simulate environmental noise and vibrations.	No false triggers; stable detection threshold maintained.	

4. IMU (MPU6050)

Test No.	Procedure	Expected Result	Observed Result / Remarks
11.	Read sensor values using I^2C connection.	Acceleration and gyroscope data change with orientation.	
12.	Test filtered data.	Stable and smooth pitch/roll readings.	

C. Actuation Modules

1. Motor Driver (BTS7960) and DC Motors

Test No.	Procedure	Expected Result	Observed Result / Remarks
16.	Apply PWM signals to motor driver without load.	Correct voltage levels at output terminals.	
17.	Connect single motor and test forward/reverse motion.	Motor rotates correctly with direction control.	
18.	Connect both motors per side in parallel.	Equal speed and torque on both sides.	
19.	Full mobility test on flat surface.	Smooth forward, reverse, and turning motion.	

D. Communication and Control

ESP32 and ESP32-CAM

Test No.	Procedure	Expected Result	Observed Result / Remarks

20.	Power and boot ESP32-CAM via FTDI.	Successful boot log on Serial Monitor.	
21.	Capture image and save to SD card.	Image file successfully created.	

Wi-Fi and Mobile App Interface

Test No.	Procedure	Expected Result	Observed Result / Remarks
22.	Connect mobile device to ESP32 access point.	Device connected; IP address assigned.	
23.	Test command endpoints (/status, /control).	Correct response from ESP32; robot obeys commands.	

E. Machine Learning (Algorithmic Function)

Test No.	Procedure	Expected Result	Observed Result / Remarks
24.	Load ML model and run inference on test data.	Returns valid probability value (0–1 range).	

25.	Feed both safe and risky scenarios.	Correct classification by the model.	

F. Integration and System Tests

Test No.	Procedure	Expected Result	Observed Result / Remarks
26.	Run complete robot operation (patrolling, obstacle avoidance, detection, logging).	All modules function in coordination without failure.	
27.	Continuous runtime test.	Stable performance; no overheating or system crash.	
28.	Emergency stop test using command or switch.	Immediate halt of motors; system enters safe state.	

6.3 Test Log Template

Test ID	Module Name	Test Performed By	Date	Expected Result	Observed Result	Remarks

---

6.4 Acceptance Criteria

A test will be marked as *successful* if:

- The module performs its intended function under defined conditions.
- Voltage, current, and thermal readings remain within safe operating limits.
- The robot correctly identifies metallic objects and responds with appropriate actions.
- Safety mechanisms such as BMS cutoff and emergency stop function correctly.
- The robot's communication with the mobile application remains stable throughout operation.

Final Test Status: _____
Date: _____

7. Implementation & Maintenance

7.1 Implementation

The implementation phase focuses on the assembly, wiring, programming, and integration of all hardware and software components into a functional prototype.
For this project, the **Land Mine Detection Robot using ESP32** was implemented in multiple stages to ensure modular development and easier debugging.

A. Hardware Implementation

1. **Mechanical Assembly:**
   The robot frame was designed using a **dual-layer metallic chassis** for rigidity and vibration isolation. The lower layer housed the motors, wheels, and motor drivers, while the upper layer contained the sensors, ESP32 controller, and power modules.

2. **Electronic Interfacing:**
   All sensors — including **metal detector**, **ultrasonic modules**, **MPU6050 IMU**, and **INA219 power monitor** — were interfaced directly to the ESP32 microcontroller.
   The system used a **3S Li-ion battery (11.1V)** with a **20A BMS** and **LM2596 buck converter** to regulate 5V power for all peripherals.

3. **Signal Integration:**
   Sensor outputs were connected via analog and digital GPIO pins using **I^2C**, **SPI**, and **UART** protocols to ensure synchronized data acquisition. Proper grounding and shielding were used to minimize noise and signal interference.

4. **Actuator Setup:**
   The robot used **two BTS7960 motor driver modules** to control

four 300 RPM DC motors for precise motion and differential steering. A **servo motor (SG90)** was mounted to rotate the metal detection coil.

B. Software Implementation

The control software was written in **Embedded C/C++** using **Arduino IDE**.

- Sensor libraries (INA219, MPU6050, Servo, WiFi, SD) were integrated for modularity.
- The logic flow was designed based on the **flowchart and DFD** described earlier.
- Machine Learning algorithms were implemented using **TensorFlow Lite Micro**, optimized for the ESP32 hardware.
- A **custom Android application** was developed using **Android Studio** for live monitoring, manual control, and image viewing.

C. Integration Testing

After all modules were individually tested, they were integrated into the final system. The robot was tested in different surface and light conditions to evaluate metal detection accuracy, mobility, and Wi-Fi communication reliability.

7.2 Maintenance

Maintenance is critical to ensure long-term performance and reliability of the robot. The following preventive measures were defined:

1. **Battery Maintenance:**
   Charge the 3S Li-ion battery only with compatible chargers. Avoid deep discharge below 9V.

2. **Periodic Sensor Calibration:**
   Calibrate the **MPU6050** and **metal detector** every 30 operating hours for accuracy.
3. **Firmware Updates:**
   Use Arduino IDE or OTA updates to modify firmware or improve ML models.
4. **Mechanical Servicing:**
   Regularly tighten screws, lubricate motor shafts, and check servo mounts for wear.
5. **Dust Protection:**
   Clean sensor openings and ensure SD card slot is sealed when operating in dusty environments.
6. **Safety Protocol:**
   Always verify wiring before powering the circuit; check voltage levels using the INA219 module.

8. Snapshots

Below are the implementation and testing phase snapshots of the project (insert your photos on this page):

8.1 Hardware Assembly

- Robot Chassis with Dual-Layer Structure
- Wiring of ESP32, Motor Drivers, and Sensors
- Power Distribution using LM2596 and 3S Li-ion Battery

8.2 Software & Interface

- Arduino IDE: Code Upload & Serial Output
- Android App Dashboard Showing Live Feed
- SD Card Log Files and Captured Images

9. Conclusion

The project titled **“Land Mine Detection Robot using ESP32”** was successfully designed, implemented, and tested to demonstrate the potential of intelligent robotics in defense and humanitarian applications.
The main objective of this project was to design a semi-autonomous mobile robot capable of detecting buried metallic landmines while ensuring the safety of human operators. Through systematic design, careful hardware selection, and precise control algorithms, the objectives of the project have been effectively achieved.

The robot integrates **mechanical engineering principles** with **embedded electronics and artificial intelligence**, resulting in a hybrid system capable of both autonomous and manual operations. The chassis has been constructed to provide structural stability, load balance, and vibration resistance on uneven terrains.
All major hardware components — including the **ESP32 microcontroller**, **ultrasonic sensors**, **BTS7960 motor drivers**, **INA219 power monitoring unit**, and **MPU6050 IMU** — have been successfully interfaced to ensure seamless data acquisition and actuation control.

The implementation of **machine learning-based pattern recognition** further enhances the robot’s intelligence, allowing it to classify the surrounding environment and predict possible mine-affected areas with better accuracy.
The addition of a **custom-developed mobile application** for live monitoring, control, and image retrieval has significantly improved user accessibility and situational awareness in field operations. The robot’s ability to **capture and store images** of detected regions ensures reliable post-mission analysis and data validation.

From a performance standpoint, the system has proven to be efficient, stable, and adaptable. The modular design approach has simplified troubleshooting and future upgradation possibilities. The robot's operation on a **rechargeable 3S Li-ion battery pack** demonstrates portability and energy efficiency, while its communication through Wi-Fi ensures convenient remote accessibility.

This project also highlights the potential of integrating **mechanical systems with intelligent control** to create field-deployable autonomous machines. It demonstrates how low-cost components, when integrated with effective software algorithms, can yield advanced functionality comparable to expensive commercial systems.

In a broader sense, this project not only contributes to the advancement of **robotic mine detection technologies** but also provides a strong educational foundation for future research in **mechatronics, AI-driven control systems, and autonomous robotics**. The experience gained during this project offers valuable insights into interdisciplinary integration, real-world problem solving, and practical system optimization.

In conclusion, the **Land Mine Detection Robot using ESP32** represents a successful prototype of an intelligent, cost-effective, and reliable system for the detection of landmines in hazardous zones. With further development and incorporation of advanced technologies such as **LiDAR, Ground Penetrating Radar (GPR), GPS navigation, and high-performance AI modules**, this system can be transformed into a fully autonomous and field-ready defense solution in the near future.

10. Future Scope

The current prototype of the **Land Mine Detection Robot using ESP32** represents a major step toward safer, low-cost, and intelligent mine detection operations. However, this project serves as a foundation upon which several advanced technologies can be integrated to further improve **accuracy, reliability, autonomy, and real-world applicability**.
Future enhancements can transform this prototype into a fully deployable **defense-grade autonomous unmanned ground vehicle (UGV)** for demining and explosive ordnance disposal (EOD) missions.

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1. Advanced Sensor Integration

To enhance the detection range, precision, and material discrimination capability, high-end sensors can be incorporated into the system:

- **Ground Penetrating Radar (GPR):** Enables the detection of deeply buried or non-metallic explosive materials by analyzing subsurface reflections. It can operate in conjunction with the metal detector for dual-sensor verification.
- **LiDAR (Light Detection and Ranging):** Facilitates high-resolution 3D mapping, obstacle profiling, and terrain reconstruction to support autonomous navigation and route optimization.
- **Hyperspectral and Multispectral Imaging Sensors:** Provide chemical and mineral composition analysis of soil to identify explosive compounds and disturbed ground surfaces indicative of buried landmines.
- **Infrared (IR) and Thermal Imaging Sensors:** Detect temperature anomalies and residual heat signatures of recently deployed mines or active explosives.

- **Millimeter-Wave Radar:** Offers high-penetration depth for subsurface imaging with centimeter-level accuracy, effective in identifying non-metallic and plastic-cased explosives.
- **X-Band and C-Band Synthetic Aperture Radar (SAR):** Enables wide-area minefield scanning with high spatial resolution, even through dense vegetation or dust.

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2. Artificial Intelligence and Data Analytics

Integrating **Artificial Intelligence (AI)** and **Machine Learning (ML)** can make the robot more autonomous, adaptive, and intelligent in its operation.

- **Deep Neural Networks (DNN) and Convolutional Neural Networks (CNN):** Can be trained for object recognition, mine-type classification, and false alarm reduction through camera imagery and sensor fusion data.
- **Reinforcement Learning (RL):** Can enable autonomous path planning and adaptive terrain navigation without prior mapping.
- **Fuzzy Logic and Adaptive Decision Systems:** For real-time decision-making in uncertain or noisy environments.
- **Edge-AI Inference Systems:** Using high-performance boards like **NVIDIA Jetson Orin Nano**, **Rockchip RK3588**, or **Khadas VIM4** for real-time ML computations without relying on cloud servers.
- **Cloud-based Collaborative Learning:** For multi-robot data sharing, dynamic re-training of models, and terrain pattern recognition across global minefield datasets.

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3. Mechanical and Structural Enhancements

From a mechanical engineering perspective, future versions can include:

- **Shock-Resistant All-Terrain Chassis:** Equipped with independent suspension and dynamic load-balancing for stability in rugged terrains and uneven surfaces.
- **Carbon-Fiber and Kevlar Composite Body:** To make the robot lightweight, durable, and explosion-resistant.
- **Sealed IP67 Enclosure Design:** To make the system waterproof and dustproof for military-grade deployment.
- **Servo-Assisted Arm or Manipulator:** To mark or excavate detected mines for manual disposal teams.

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4. Power and Energy Optimization

Energy management will play a critical role in ensuring longer mission endurance:

- **Solar-Assisted Hybrid Power System:** Integration of flexible solar panels on the chassis for passive charging during daytime patrols.
- **Smart Battery Management System (BMS):** For real-time health monitoring, predictive maintenance, and thermal control of high-capacity Li-ion packs.
- **Supercapacitor Modules:** To handle transient current surges during high-load conditions like acceleration or servo movement.
- **Hydrogen Fuel Cell Integration:** For extended mission duration and faster refueling capability.

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5. Communication, Networking, and Control

Enhanced communication architecture can make the robot part of a larger coordinated defense network:

- **CAN Bus / RS485 Industrial Protocols:** For reliable internal communication among modules.
- **4G / 5G Cellular Modules:** For remote operation and data transmission across defense control networks.
- **Mesh Networking:** To enable inter-robot communication in multi-agent mine detection missions.
- **Satellite Connectivity (SBAS / GNSS):** For precise GPS correction and global area operations in remote zones.
- **Encrypted Communication Protocols:** For secure command and control in defense missions.

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6. System-Level and Autonomy Upgrades

- **SLAM (Simultaneous Localization and Mapping):** For real-time mapping and autonomous navigation in unknown environments using LiDAR and vision-based algorithms.
- **GPS-Guided Autonomous Route Planning:** For systematic area coverage and path optimization during minefield scanning.
- **Swarm Robotics Coordination:** Multiple robots can communicate and coordinate to cover large areas efficiently using AI-based task distribution.
- **Integration with UAVs (Drones):** For overhead mapping, terrain scanning, and visual tracking of ground robots during missions.
- **Integration with UGV–UAV Hybrid Systems:** For combined ground and aerial surveillance and detection missions.

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7. Mine Disposal and Neutralization Capabilities

Future versions of the system can go beyond detection and include **automated disposal mechanisms**:

- **Robotic Arm with Explosive Neutralization Tool (SBC - Smart Bomb Cutter):** To safely defuse or expose detected landmines.
- **Directional Water Jet or Pyro-Disruptor System:** For remote mine detonation or neutralization using high-pressure jets or controlled explosive charges.
- **Thermal and Microwave Neutralization Units:** To deactivate or melt explosive material from a safe distance.
- **Autonomous Marking Mechanism:** Using paint or GPS coordinates to mark detected mine positions for follow-up disposal teams.

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8. Integration with Defense and Humanitarian Operations

- The robot can be connected to **Command and Control Centers (C4ISR)** for real-time operation monitoring.
- Integration with **AI-driven battlefield management systems** for predictive risk mapping and minefield analytics.
- Use in **post-conflict humanitarian missions** for large-scale clearance of mine-affected civilian regions.
- Collaboration with **UN Mine Action Service (UNMAS)** and **DRDO** for field testing and validation of autonomous mine detection prototypes.

Summary

With the integration of the above technologies, the **Land Mine Detection Robot** can evolve from a prototype into a **fully autonomous, intelligent, and multi-sensor demining platform**.
By combining **AI-based data fusion**, **high-precision mapping**, **adaptive power systems**, and **automated neutralization capabilities**, future versions of this system could play a crucial role in global efforts toward **mine-free zones and next-generation battlefield automation**.

11. Appendices

Appendix A – Component Specifications

Component	Specification Summary
ESP32 DevKit V1 (WROOM-32)	Dual-core MCU (Xtensa 240 MHz), 30+ GPIOs, Wi-Fi + BLE, 3.3V logic
ESP32-CAM (AI Thinker)	2 MP OV2640 Camera, SD Card Slot, 5V input, UART/Serial communication
HC-SR04 Ultrasonic Sensor (×4)	Range: 2–400 cm, 40 kHz operation, 5V logic, trigger–echo interface
BTS7960 Motor Driver (×2)	43A Dual H-Bridge driver, PWM compatible, 5–27V operation
300 RPM DC Motors (×4)	6–12V operation, dual-shaft, geared type
INA219 Power Monitor	Voltage range: 0–26V, Current: ±3.2A, I^2C interface
MPU6050 IMU Sensor	3-axis accelerometer + 3-axis gyroscope, I^2C protocol
Metal Detector Module (LC type)	Pulse Induction-based coil detector, adjustable sensitivity
SG90 Servo Motor	180° micro servo, torque: 2.5 kg·cm, 5V operation
LM2596 Buck Converter	Adjustable 4.5–35V input, 1.25–30V output, 3A max current
3S Li-ion Battery Pack	11.1V nominal, 12.6V max, protected by 3S 20A BMS

Component	Specification Summary
Buzzer Module	5V active buzzer for alert signaling
SD Card Module	SPI interface, 3.3V logic compatible
Connecting Accessories	Male-female jumper wires, XT60/XT30 connectors, screw terminals, chassis mount kit

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Appendix B – Pin Configuration and Wiring Table

ESP32 DevKit V1 Wiring Summary

Module / Component	Signal	ESP32 Pin	Voltage (V)	Notes
Metal Detector	Output Signal	GPIO34	3.3V logic	Shared through buzzer for detection feedback
	VCC	5V	From LM2596 output	
	GND	GND	Common ground	
IMU (MPU6050)	SDA	GPIO21	3.3V	I^2C data
	SCL	GPIO22	3.3V	I^2C clock
	VCC	3.3V	---	
	GND	GND	---	
Ultrasonic Sensors (4 Units)	Front TRIG	GPIO5	5V	Forward sensor

	Front ECHO	GPIO18	3.3V	
	Left TRIG	GPIO4	5V	Left sensor
	Left ECHO	GPIO2	3.3V	
	Right TRIG	GPIO15	5V	Right sensor
	Right ECHO	GPIO13	3.3V	
	Rear TRIG	GPIO12	5V	Rear sensor
	Rear ECHO	GPIO14	3.3V	
Motor Driver 1 (Left Side)	RPWM	GPIO26	5V	Left motor forward
	LPWM	GPIO27	5V	Left motor reverse
Motor Driver 2 (Right Side)	RPWM	GPIO25	5V	Right motor forward
	LPWM	GPIO32	5V	Right motor reverse
Buzzer (Alert)	Signal	GPIO33	5V	Active buzzer
	GND	GND	---	
INA219 Power Sensor	SDA	GPIO21	3.3V	Shared I²C bus

	SCL	GPIO22	3.3V	Shared I²C bus
	VCC	3.3V	---	
	GND	GND	---	
ESP32-CAM Communication Link	UOR(RX)	GPIO17	3.3V	Serial receive from CAM
	UOT(TX)	GPIO16	3.3V	Serial transmit to CAM
	5V	5V	From LM2596	
	GND	GND	---	Common ground
LM2596 Buck Converter	Input	Battery (11.1V)	---	From 3S battery
	Output	5V line	---	Powers ESP32, sensors, drivers
Battery Input	XT60/XT30 connector	---	11.1V DC	Connected to LM2596 & Motor Drivers

Power Distribution Overview

Voltage Line	Used Components	Source
5V	ESP32 DevKit, ESP32-CAM, Ultrasonic Sensors, Servo, Motor Driver Inputs, Buzzer	LM2596 Output
3.3V	IMU (MPU6050), INA219	ESP32 onboard regulator
11.1V	Motors, LM2596 Input, Battery charging terminals	3S Li-ion Battery Pack

Appendix C – Connection Summary Diagram (Description)

The wiring diagram connects all modules in a **star topology** with a **common ground**.

- The **ESP32 DevKit** acts as the central controller.
- **ESP32-CAM** is connected via UART (GPIO16–17) for image and data communication.
- All ultrasonic sensors are connected to individual trigger and echo pins for obstacle detection.
- The **IMU (MPU6050)** and **INA219** share the same I^2C bus (GPIO21, GPIO22).
- Two **BTS7960 motor drivers** drive the left and right wheel sets respectively.
- The **metal detector output** is fed through GPIO34 as a digital input for mine detection events.
- **Power distribution** is managed by LM2596 (step-down) and a 3S Li-ion pack with BMS for protection.

The design ensures:

- Stable 5V power supply lines from LM2596.
- Common ground reference across all modules.
- Minimal interference between analog and digital signals by proper wire separation.
- Reusable headers and connectors for maintenance and replacement.

Appendix D – Code Structure Overview

The software is divided into functional modules for maintainability and debugging efficiency:

File Name	Purpose
main.ino	Initializes all peripherals, runs main control loop.
sensors.cpp / sensors.h	Functions for ultrasonic, IMU, INA219, and metal detector readings.
motors.cpp / motors.h	Controls motor drivers, direction, and speed using PWM.
alerts.cpp / alerts.h	Manages buzzer and LED visual/audio feedback.
comms.cpp / comms.h	Handles Wi-Fi communication, app data, and SD card interface.
ml_inference.cpp / ml_inference.h	Implements ML model logic and probability scoring.

Appendix E – Testing Documentation (Summary)

During the hardware validation phase, the following aspects were verified:

- Stability of 5V line under dynamic load.
- Reliable SPI and I^2C communication.
- Accurate distance readings from ultrasonic sensors.
- Correct motion control from both BTS7960 modules.
- Proper synchronization between ESP32 and ESP32-CAM via UART.

All modules were calibrated and configured for noise-free operation before integration.

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